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14 June 1982
SUPERSEDING
MIL-M-10891B(MU)
15 February 1973

MILITARY SPECIFICATION

METHYLPHOSPHONIC DICHLORIDE, TECHNICAL

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of methylphosphonic dichloride, CH_3POCl_2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Ammunition Lot Numbering

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Methylphosphonic dichloride shall conform to the chemical and physical characteristics of table I when tested as specified therein.

3.2 Preproduction. Prior to the start of regular production, a preproduction lot of methylphosphonic dichloride shall be produced in accordance with this specification for test (see 4.3).

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TABLE I. Chemical and physical characteristics

Characteristic	Requirement		Test paragraph
	Minimum	Maximum	
Purity, mole percent:			
by freezing point	98.0		4.4.4.1
by NMR	98.0		4.4.4.2
Methylphosphonic dichloride (CH_3POCl_2), percent by weight	98.0		4.4.4.3
Sulfur, percent by weight*		0.01	4.4.4.4

*This requirement shall be applicable only to material produced when sulfur-containing compounds are used in the process.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of methylphosphonic dichloride shall be produced using the same methods, materials, equipment, and processes of manufacture as will be used during regular production. The size of the preproduction lot shall be as specified in the contract or purchase order (see 6.1). The preproduction lot shall be identified and controlled in accordance with MIL-STD-1168.

4.3.2 Sampling. Sampling of the preproduction lot and the manner of forwarding the samples to a test facility shall be as specified in the contract or purchase order (see 6.1).

4.3.3 Inspection procedure for test. The preproduction samples shall be tested as specified in 4.4.4. Failure of any test by any sample shall be cause for rejection of the preproduction lot.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the methylphosphonic dichloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.2). When shipments are by tank car, each tank car shall constitute a lot. Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.4.2 Sampling.

4.4.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test (see 6.3).

(a) Drums. Sampling shall be conducted in accordance with table II. The sample drums shall be heated in a hot air chamber having a temperature of approximately 43°C until the contents are completely liquefied. The contents shall be agitated until thoroughly mixed and then by means of a corrosive-resistant sampling thief, a 1-quart specimen shall be removed from each sample drum and quickly placed in a separate clean, dry, glass container labeled to identify the lot and the drum from which the specimen was taken.

TABLE II. Sampling of drums for tests

Number of containers in batch or lot	Number of sample containers
2 to 15	2
16 to 50	3
51 to 150	5
151 to 500	8
501 to 3,200	13
Over 3,200	20

(b) Tank cars. The contents of the tank car shall be heated to a temperature of approximately 43°C using the tank car heating coils. After the contents have been completely liquefied, a 1-quart representative specimen shall be removed in such a manner that the top, middle, and bottom layers of the methylphosphonic dichloride are approximately equally represented in the specimen. The specimen shall be quickly placed in a clean, dry, glass container labeled to identify the lot represented.

4.4.3 Inspection procedure.

4.4.3.1 For examination of packaging. The sample unit shall be one filled drum or tank car, as applicable, ready for shipment. Sample containers shall be examined for the following defects using an AQL of 2.5 percent defective:

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(a) Contents per container not as specified
(b) Container not as specified
(c) Container closures, plugs, or gaskets not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing or illegible

4.4.3.2 For test. Each sample specimen taken in 4.4.2.2 shall be tested as specified in 4.4.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.4.4 Tests (see 6.3). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. All specimens shall be liquefied and thoroughly agitated before test portions are removed from the glass containers. Tests shall be conducted as follows:

4.4.4.1 Purity by freezing point.

(a) Procedure. Use a freezing point apparatus similar to that shown in figure 1. Dry the apparatus in a vacuum oven at 130° ± 5°C for 6 hours to remove all of the moisture. To aid crystal formation, place a 0.25 to 0.5 millimeter (mm) diameter piece of dry charcoal in the inner chamber of the apparatus. Immediately transfer the apparatus to a dry glove box filled with a nitrogen atmosphere and containing a bottle of the specimen. Do not allow the apparatus to cool to ambient temperature. Add enough specimen to the inner chamber of the apparatus to cover the temperature sensitive portion of the temperature probe by approximately 1 centimeter (cm). Attach the stirring assembly and seal the shaft of the stirrer so that the inner chamber is sealed from the atmosphere. Remove the apparatus from the glove box and attach the stirring assembly to a variable speed stirring motor. Insert into the thermowell a device such as a resistance thermometer or thermocouple which is capable of measuring temperature to the nearest 0.01°C. Fill the thermowell with ethylene glycol for better heat conduction. Connect the temperature probe to a temperature recorder capable of recording to the nearest 0.01°C. Apply a vacuum of 2 mm of mercury to the outer chamber. Place the freezing point apparatus in an ice-water bath and turn on the stirring motor. Record the cooling of the specimen as temperature versus time. The curve should consist of a steeply sloped line and an aftercurve produced during crystallization. The freezing point of the specimen is the intersection of the steeply sloped cooling portion of the curve and the extrapolation of the aftercurve. Remove the ice-water bath and replace it with a Dewar flask containing 43°C water. Record the melting curve of the specimen as temperature versus time.

(b) Calculation. The freezing point of pure methylphosphonic dichloride is taken as 32.74°C (305.89°K). Calculate the mole percent purity as follows:

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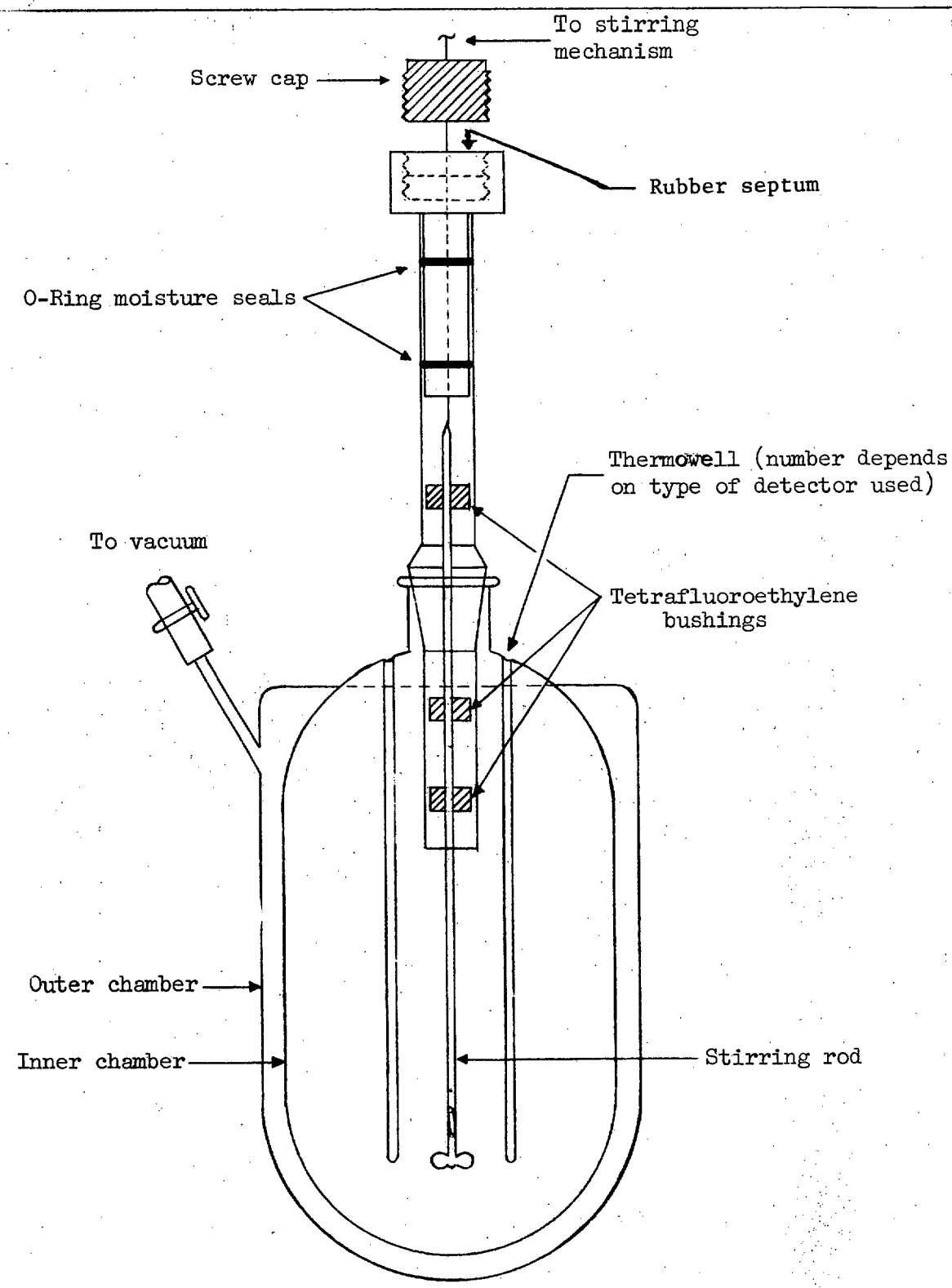


Figure 1. Typical Freezing Point Apparatus

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$$\log X = 2 - 1051.12 \left(\frac{1}{T} - \frac{1}{305.89}\right)$$

Mole percent purity = X

where: X = Mole percent purity of methylphosphonic dichloride,
log X = Common logarithm of X, and
T = Observed freezing point, °K.

4.4.4.2 Purity by nuclear magnetic resonance (NMR) spectroscopy.

4.4.4.2.1 ^1H NMR characterization.

(a) Spectrometer. Use a spectrometer which has a minimum magnetic field strength of 14 kilogauss [^1H, 60 megahertz (MHz)] and a suitable internal or external field/frequency lock. The spectrometer should have a resolution of 0.6 hertz (Hz) or less. Resolution is defined as the line width at half-height of a single peak such as tetramethylsilane (TMS) in deuterated chloroform. The spectrometer should be capable of sweeping from 0 parts per million (ppm), TMS to at least a downfield value of 17 ppm. The sensitivity of the spectrometer should be such that the signal-to-noise ratio (S/N) is at least 15 to 1 for a 1-percent solution by weight of ethyl benzene in carbon tetrachloride (CCl_4). The S/N is defined as follows:

$$S/N = \frac{2.5\ A}{B}$$

where: A = Height of tallest peak of ethyl benzene quartet, and
B = Height of baseline noise (peak to peak).

The electronic integral of the spectrometer should have at least a 1-percent electronic accuracy in the integrating circuit. Also, for the total integral of the three groups in the spectrum of a 5-percent solution by volume of ethyl benzene in CCl_4, the average deviation of the integrals will not exceed $\pm$ 2 percent for 5 successive scans. For spectrometers that do not have 1-percent electronic accuracy in the integrating circuit but do have 1-percent accuracy in the spectrum amplitude controls, the spectrum amplitude control may be used to increase the integral displacement as described in the procedures below. The integral amplitude control must not be reset after obtaining the initial displacement of the integral in this type of spectrometer.

(b) Preparation of specimen. Dry the NMR tubes and any glassware to be used in a vacuum oven at 130 $\pm$ 5°C for 6 hours to remove all moisture. Store the NMR pressure caps in a desiccator over phosphorus pentoxide prior to use. In a dry glove box or glove bag filled with nitrogen atmosphere, transfer 0.4 to 0.5 milliliter (ml) of the liquid specimen into a clean, dry, 5-mm "Pyrex" or equal NMR tube and seal the tube with a pressure cap. (If the specimen to be characterized is a solid, gently warm the container until the specimen completely liquefies, and agitate gently until the contents are thoroughly

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mixed.) Wrap the top of the capped NMR tube with "Parafilm" or equal. Do not add TMS or any other standard reference material to the specimen as contamination may result. Make sure that there are no solid particles, including dust, in the NMR tube as this could interfere with the analysis by upsetting the homogeneity.

(c) Procedure. Place the NMR tube containing the specimen into the probe of the spectrometer and turn on the spinner. If the specimen being tested is a solid, increase the probe temperature to approximately 40°C. Allow thermal equilibration of the sample before beginning the NMR test. Use a 500-Hz sweep width or a value as close to 500 Hz as possible. Center the methylphosphonic dichloride resonance at delta 2.56 ppm on the chart paper and tune the instrument for maximum homogeneity. Select an amplitude setting such that the peaks are 90 to 95 percent of the height of the chart paper. Run the spectrum from 17 ppm to 0 ppm (TMS) at a sweep time of approximately 250 seconds, offsetting the sweep as required. Observe the ^{1}H NMR characteristics. Methylphosphonic dichloride is a doublet, delta 2.56, with a proton-to-phosphorus coupling constant (J_{P-H}) of 16.6 $\pm$ 0.2 Hz. If the resonances appear to be broadened or the normal "ringing pattern" is not observed in the spectrum, check the spectrometer homogeneity using a sample of TMS in chloroform. Proper homogeneity is evidenced by a normal "ringing pattern" and a line width of 0.6 Hz or less (peak width at half-height) for the TMS peak. Improper homogeneity with this TMS sample indicates the need to retune the instrument. If the broadening is not of instrumental origin, it must be assumed that the sample contains paramagnetic impurities (i.e., metal ions). If the line widths are so broad that the resonance of the methylphosphonic dichloride cannot be recognized as a doublet, confirmation of structure should be based on the specimen's ^{31}P NMR characteristics. Replot the spectrum from 17 ppm to 0 ppm (TMS) at the maximum amplitude of the spectrometer, keeping the sweep width at approximately 500 Hz and the sweep time at approximately 250 seconds, offsetting the sweep as required. Determine any spinning sidebands in the spectrum by varying the spinning speed of the specimen tube and replotting the spectrum.

(d) Determination of impurities. From the replotted spectrum above, determine which peaks are due to carbon-13 sidebands, spinning sidebands, and instrumental sidebands. Those peaks which cannot be assigned as sidebands are impurity peaks. If no peaks other than those attributable to methylphosphonic dichloride and its sidebands are observed, proceed to 4.4.4.2.2. Two procedures are used to quantitate the impurities observed in the ^{1}H NMR spectrum. To quantitate any impurity peaks that are 50 Hz or more from the main methylphosphonic dichloride resonance, use procedure (1) described below. To quantitate any impurity peaks less than 50 Hz from the main resonance, use procedure (2). Some samples may require the use of both (1) and (2).

(1) See figure 2. Obtain the electronic integral over the methylphosphonic dichloride resonance, making sure to include all sidebands within the integral. The integral displacement should be 70 to 90 percent of the height of the chart paper. Integrate the impurity peak(s). If the impurity peak(s) is/are small, the integral displacement will be negligible.

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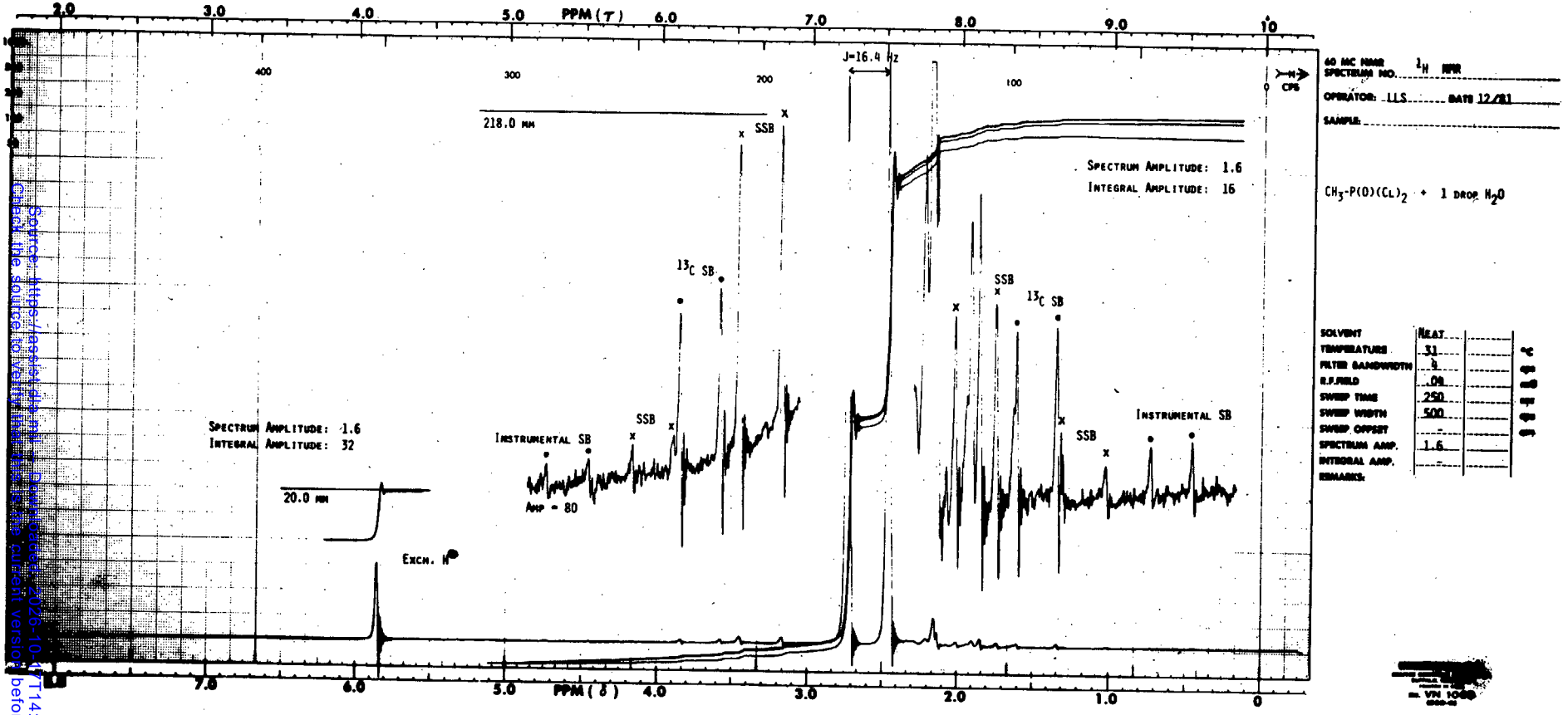


Figure 2. ^{1}H NMR Spectrum of Methylphosphonic Dichloride, 500 Hz Sweep Width

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Increase the integral displacement for the impurity peak(s) using only the integral amplitude control and/or the spectrum amplitude control. The integral displacement should be increased to at least 10 mm for each impurity peak observed. Take at least three replicate integrals for all of the resonance peaks. These replicate integrals should lie within 2 to 3 mm of each other. Note on the spectrum all integral amplitude values and spectrum amplitude values that were used.

(2) See figure 3. Using a sweep width of approximately 100 Hz, center the methylphosphonic dichloride doublet on the chart paper making sure that any impurity peaks to be integrated are also on the chart paper. Plot this spectrum at a sweep time of approximately 250 seconds using an amplitude setting such that the peaks of the methylphosphonic dichloride are approximately 100 percent of the height of the chart paper. Using the electronic integrator of the spectrometer, integrate this spectrum using a sweep time of approximately 100 seconds and an integral amplitude such that displacement over the impurity peaks is at least 2 to 3 mm. The integral may have to be taken in a step-wise fashion, overlapping scans, since the total integral may be well off of the chart paper.

4.4.4.2.2 ^{31}P NMR Characterization.

(a) Spectrometer. Use a spectrometer which has a minimum field strength of 18 kilogauss (^{31}P, 32.2 MHz) and a suitable internal or external field/frequency lock. Either a continuous wave (CW) or a Fourier transform (FT) NMR spectrometer may be used. The spectrometer should have a resolution of 1 Hz or less. Resolution is defined as the line width at half-height of a single peak such as 85-percent phosphoric acid, (H_3PO_4). The sensitivity of the spectrometer shall be such that the S/N is at least 60 to 1 for a 10-percent sample of trimethyl phosphite, $(CH_3O)_3P$, in deuterated benzene, undecoupled. The S/N is defined as follows:

$$S/N = \frac{2.5\,A}{B}$$

where: A = Height of tallest peak of $(CH_3O)_3P$ multipet, and
B = Height of baseline noise (peak to peak).

The electronic integral of the spectrometer shall have at least a 1-percent electronic accuracy in the integrating circuit. Also, the average deviation of the integrals for a sample of 5-percent H_3PO_4 will not exceed ± 2 percent for 5 successive scans. The spectrometer should be capable of sweeping from 50 ppm upfield of 85-percent H_3PO_4 (delta 0.0) to 250 ppm downfield of 85-percent H_3PO_4.

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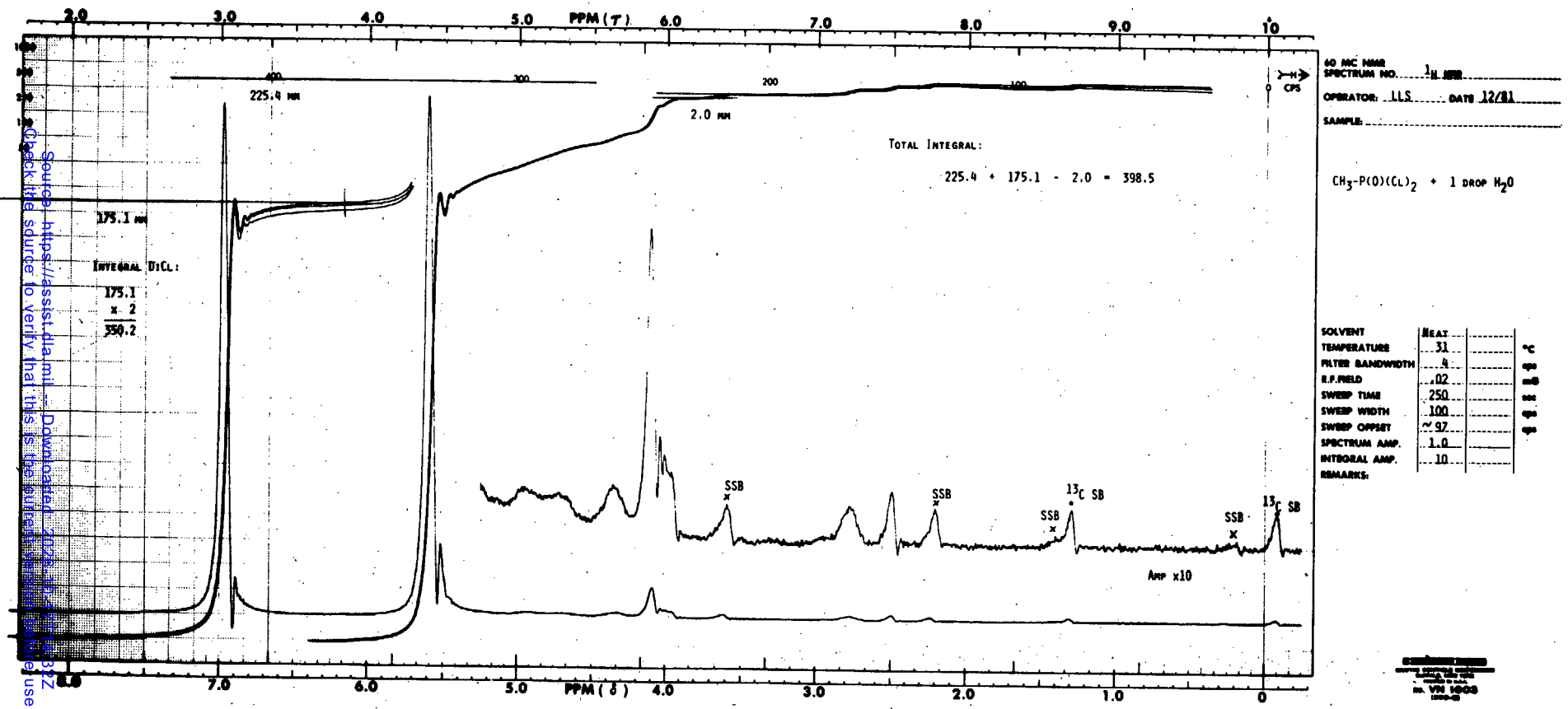


Figure 3. Expanded ^{1}H NMR Spectrum of Methylphosphonic Dichloride, 100 Hz Sweep Width

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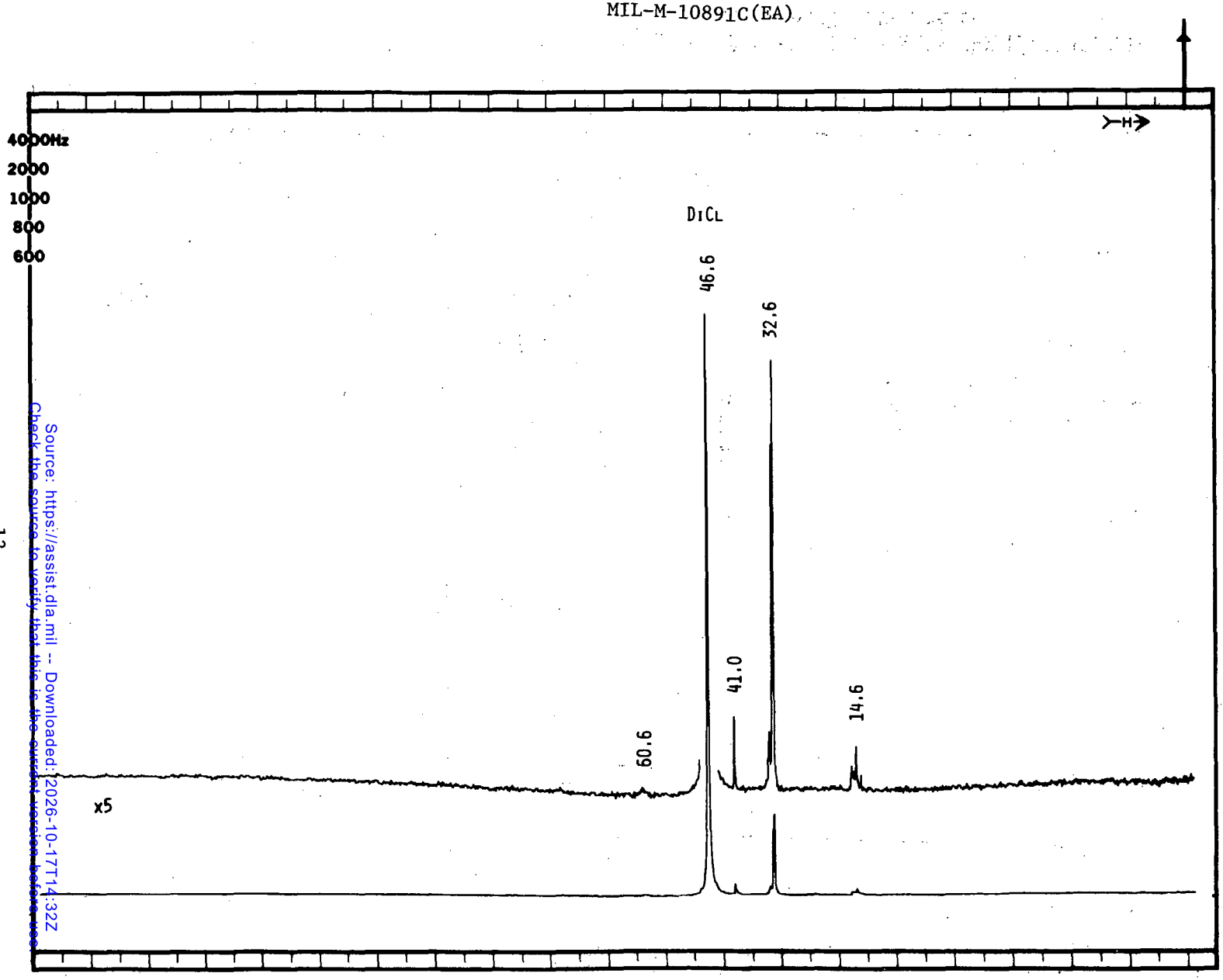
(b) Preparation of specimen. Use the specimen prepared in 4.4.4.2.1(b), if possible. If a larger NMR tube is required in order to obtain the ^{31}P NMR spectrum, prepare the specimen as described in 4.4.4.2.1(b), using the appropriate amount of specimen. If a lock signal is required, use a sealed capillary filled with the locking compound. Store the sealed capillary in a desiccator over phosphorus pentoxide prior to use. Alternately, the locking compound may be added directly to the specimen if it has been determined that it will not react with the methylphosphonic dichloride and that it is free of water and alcohols.

(c) Procedure. See figure 4. Place the NMR tube containing the specimen into the probe of the spectrometer and turn on the spinner. If the sample being tested is a solid, increase the probe temperature to approximately 40°C. Allow thermal equilibration of the specimen before beginning the NMR test. Tune the instrument for maximum homogeneity and obtain the spectrum from 250 ppm downfield of 85-percent H_3PO_4 (delta 0.0) to 50 ppm upfield of 85-percent H_3PO_4. If a CW instrument is used, scan this region using an amplitude setting such that the methylphosphonic dichloride resonance is 90 to 95 percent of the height of the chart paper. If an FT instrument is used, typically 200 to 500 transients will be required to obtain a satisfactory S/N using a sweep width of 8000 Hz, a filter bandwidth of 8000 Hz, a pulse width of 9 microseconds (34°), a pulse delay of 2.5 seconds, an acquisition time of 0.5 second, and gated decoupling to eliminate nuclear Overhauser effects (NOE). Observe the ^{31}P NMR spectrum characteristics. Methylphosphonic dichloride resonates 45 $\pm$ 2 ppm downfield of 85-percent H_3PO_4 (delta 0.0). If a CW instrument is used, replot the spectrum from delta 250 to delta -50 at the maximum amplitude of the spectrometer. If an FT instrument is used, increase the amplitude of the transformed spectrum 5 to 10 times and replot the spectrum from delta 250 to delta -50.

(d) Determination of impurities. From the spectrum plotted above, determine which peaks are due to spinning sidebands and instrumental sidebands. Those peaks which cannot be assigned as sidebands are impurity peaks. To quantitate the impurities observed in the ^{31}P NMR spectrum (see figure 5) obtain the electronic (or digital) integral over the methylphosphonic dichloride resonance, making sure to include all spinning sidebands and instrumental sidebands within the integral. The integral displacement should be 70 to 90 percent of the height of the chart paper. Integrate the impurity peak(s). If the impurity peak(s) is/are small, the integral displacement will be negligible. Increase the integral displacement for the impurity peak(s) to at least 10 mm (or units) for each impurity observed. Take at least three replicate integrals for all of the resonance peaks. These replicate integrals should be within 2 to 3 mm (or units) of each other. Note on the spectrum all integral amplitude values and spectrum amplitude values that were used.

4.4.4.2.3 Calculation of mole percent purity by NMR. The calculation of a mole percent purity for the methylphosphonic dichloride sample uses both ^{31}P and the ^{1}H NMR spectra. A sample calculation for the methylphosphonic dichloride specimen shown in figures 2 through 5 is contained in the appendix. Calculate the mole percent purity by NMR as follows:

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SPECTRUM NO. ^{31}P NMR
OPERATOR LLS DATE 12/81
SAMPLE

CH_3-P(O)(CL)$_2$ + 1 DROP H_2O

LOCK SIGNAL EXT
SPIN PATE 20 rps. TEMP 30°C

ACQUISITION
SPECTRAL WIDTH (SW) 8K Hz
NO. OF TRANSIENTS (NT) 200
ACQUISITION TIME (AT) 0.5 sec.
PULSE WIDTH (PW) 9 μsec.
PULSE DELAY (PD) 2.5 sec.
DATA POINTS (DP) 8K

TRANSMITTER OFFSET (TO)
HIGH FIELD X LOW FIELD
RECEIVER GAIN (RG) 5

DECOUPLER MODE (DM) 3
DECOUPLER OFFSET (DO) 56
NOISE BANDWIDTH (NB) -2 kHz

DISPLAY
SENS. ENHANCEMENT (SE) 0.5 sec.
WIDTH OF PLOT (WP) 8K Hz
END OF PLOT (EP) 0 Hz
WIDTH OF CHART (WC) 8K Hz
END OF CHART (EC) 0 Hz
VERTICAL SCALE (VS) 100
REFERENCE LINE (RL) EXT H_3PO_4 (85%)

Figure 4. ^{31}P NMR Spectrum of Methylphosphonic Dichloride, 8000 Hz Sweep Width

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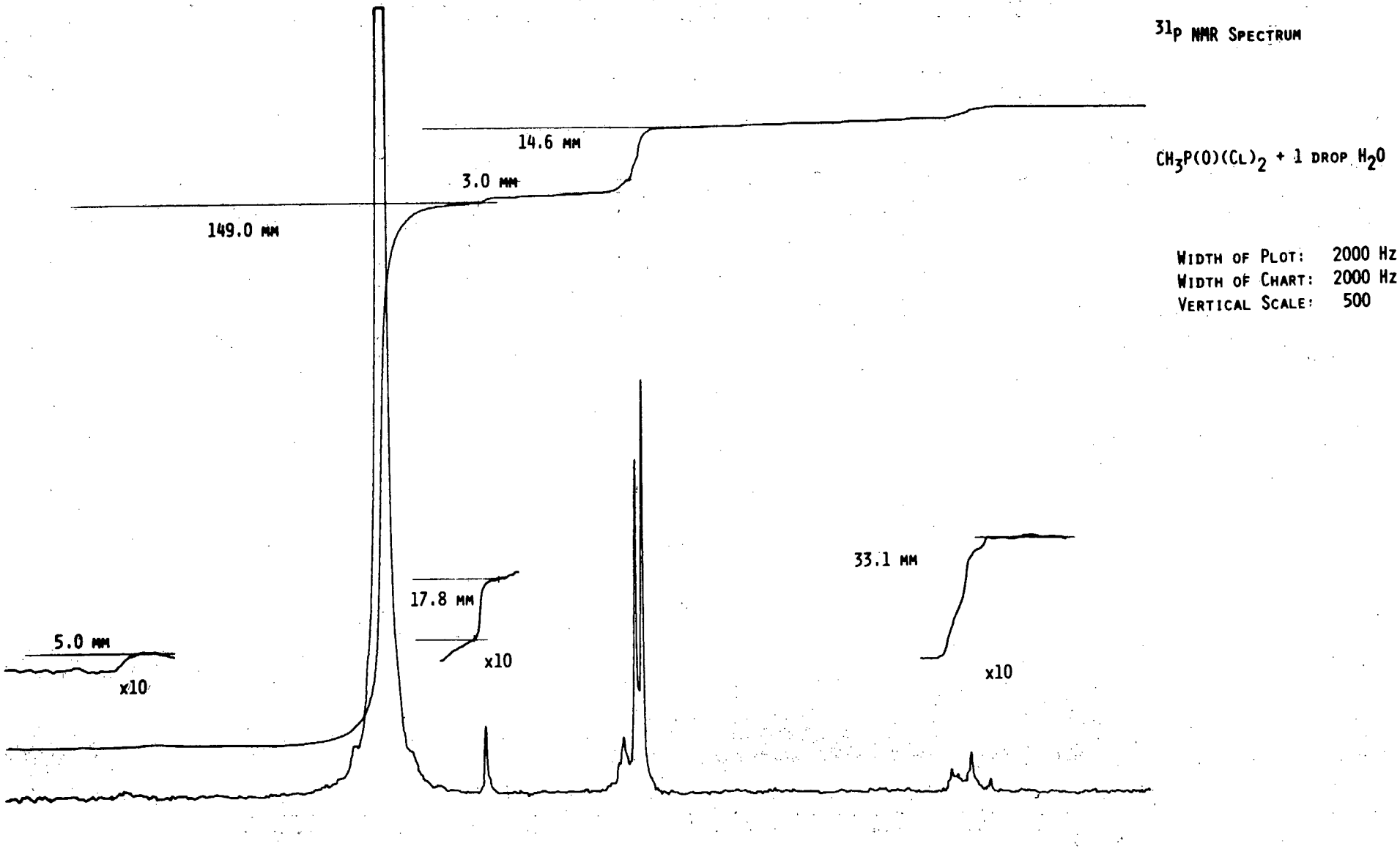


Figure 5. Expanded 31p NMR Spectrum of Methylphosphonic Dichloride, 2000 Hz Sweep Width

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(a) Observe the ^{31}P NMR spectrum and assign as many resonances as possible.

(b) Add together all the areas of the resonances containing the moiety CH_3-P, excluding the resonance for the methylphosphonic dichloride (delta 45 $\pm$ 2). Make sure that all areas are on the same scale (i.e., have the same integral amplitude and spectrum amplitude). If not, adjust all of the integral values to the same scale using direct proportions.

(c) Using the 100-Hz ^1H NMR spectrum recorded in 4.4.4.2.1(d)(2), find the area percent of the methylphosphonic dichloride for this region of the spectrum. To do this, take the integral of the methylphosphonic dichloride peaks and divide by the total integral of this region. Exclude all of the methylphosphonic dichloride sidebands.

(d) Using the 500-Hz ^1H NMR spectrum recorded in 4.4.4.2.1(d)(1), measure the integral over the methylphosphonic dichloride region (delta 0.00 to delta 5.00). Multiply this integral by the area percent calculated in (c) and then divide by three. This number is the integral value per proton for the methylphosphonic dichloride.

(e) Using the 500-Hz ^1H NMR spectrum, calculate the area percent of CH_3-P impurities by multiplying the integral value per proton for the methylphosphonic dichloride by three and subtracting this value from the total integral.

(f) Using the 500-Hz ^1H NMR spectrum, calculate the integral value per proton for each CH_3-P containing impurity. To do this, use the ^{31}P NMR data calculated in (b). For each impurity, take the area found in the ^{31}P NMR spectrum, divide by the total CH_3-P area in the ^{31}P NMR spectrum, multiply by the area of the CH_3-P impurities in the 500-Hz ^1H NMR spectrum and then divide by the number of CH_3 protons in the impurity.

(g) Using the ^{31}P NMR spectrum, determine any phosphorus-containing impurities that were not included in (a) (example: $POCl_3$, delta approximately 2). For each impurity, calculate the integral value to be used in the purity calculation in (i). Use the ratio of the impurity resonance to the methylphosphonic dichloride resonance (delta 45 $\pm$ 2) in the ^{31}P NMR spectrum to calculate what the integral value would be for the 500-Hz ^1H NMR spectrum. Use direct proportions. If the impurity resonance represents a compound containing more than one phosphorus nucleus, be sure to divide by the number of phosphorus nuclei in the compound in order to obtain the correct value.

(h) Using the 500-Hz ^1H NMR spectrum, determine any proton-containing impurities not included in (f) (examples: HCl, $CHCl_3$). Calculate the integral value per proton for each impurity. If the integral was taken at a different integral and/or spectrum amplitude setting, make sure that the values are scaled to those values determined for the impurities in (f) and to that determined for the methylphosphonic dichloride in (d). Use direct proportions. When evaluating the exchangeable proton peak, be sure to subtract

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any contribution to the ingetral made by any of the phosphorus compounds in (f) or any of the other proton-containing impurities that may have an exchangeable proton.

(i) Add together the integral value per proton for the methylphosphonic dicloride from (d) and the integral values per proton (or per mole) for each impurity determined in (f), (g), and (h). To determine the mole percent purity of the methylphosphonic dichloride, divide the integral value per proton for the methylphosphonic dichloride from (d) by the sum obtained in (i). Multiply by 100 to convert to percent.

4.4.4.3 Methylphosphonic dichloride. Place approximately 0.25 gram (g) of the specimen, weighed to the nearest milligram, in a glass sampling bulb. Place the bulb in a heavy-walled, glass-stoppered Erlenmeyer flask containing 50 ml of 10-percent sodium hydroxide solution. Break the bulb by vigorously shaking the flask and continue shaking until all vapors in the flask are absorbed. Remove the stopper and by means of a suitable rod crush the stem of the bulb. Replace the stopper and allow the solution to stand for approximately 30 minutes. Acidify with nitric acid and determine chlorine by the Volhard method. Calculate the percent by weight methylphosphonic dichloride as follows:

$$\text{Percent methylphosphonic dichloride} = \frac{6.646\ (AB - CD)}{W}$$

where: A = Milliliters of silver nitrate solution,
B = Normality of the silver nitrate solution,
C = Milliliters of potassium thiocyanate solution,
D = Normality of the potassium thiocyanate solution, and
W = Weight in grams of specimen.

4.4.4.4 Sulfur. Place 300 ± 5 g of specimen, weighed to the nearest 0.01 g, into a 500-ml distillation flask. Distill off approximately 10 g through a fractionation column 15 inches long and 1 inch in diameter packed with glass helices. Weigh the distillate. From the distillate take three samples and test in triplicate as follows: Weigh approximately 0.5 g of the distillate in a gelatine capsule and fuse in a nickel Paar bomb with a mixture containing 1 g of potassium chlorate, 0.3 g of benzoic acid, and 15 g of sodium peroxide. Ignite, cool, and leach the residue by placing the bomb on its side in a 600-ml beaker and adding sufficient water to cover half of the bomb. When the reaction subsides, remove the bomb and wash several times with approximately 2 N hydrochloric acid, catching the washings in the same beaker. Completely dissolve the residue by adding approximately 8 N hydrochloric acid saturated with bromine. Evaporate the bromine and filter the solution through filter paper into a 600-ml beaker. Neutralize the filtrate with ammonium hydroxide solution, then make slightly acid with hydrochloric acid. Heat to approximately 98°C and add an excess of 15-percent barium chloride solution in small quantities with stirring. Let stand 4 hours or preferably overnight. Filter through a tared Gooch crucible and wash with hot water until the filtrate is free of barium chloride. Ignite the crucible to dull red, cool to room temperature, weigh, and calculate the percent by weight sulfur as follows:

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$$\text{Percent sulfur} = \frac{13.73AB}{CD}$$

where: A = Weight in grams of barium sulfate,
B = Weight in grams of distillate,
C = Weight in grams of specimen before distillation, and
D = Weight in grams of specimen from distillate.

5. PACKAGING

5.1 Packing, interplant shipment. Methylphosphonic dichloride shall be packed in 55-gallon drums or tank cars as specified (see 6.1).

5.1.1 Drums. A quantity of 500 (+5 or -0) pounds of methylphosphonic dichloride shall be packed in accordance with Department of Transportation (DOT) regulations in a nominal 55-gallon capacity nickel drum conforming to DOT Specification 5K. At the time of filling, the drum shall be clean and dry. The drum shall be furnished with a 3/4-inch diameter closure and a 2-inch diameter closure. The threaded plugs for the closures shall be fitted with leak-proof gaskets formed from tetrafluoroethylene or fluorinated ethylene-propylene resins. The drum shall be closed by tightening the 3/4-inch diameter plug against the gasket to a torque of 15 to 18 pound-feet and by tightening the 2-inch plug to a torque of 30 to 35 pound-feet. The exterior of the drum shall be free of any contamination by methylphosphonic dichloride or other matter.

5.1.2 Tank cars. A quantity of 86,000 (+500 or -0) pounds of methylphosphonic dichloride shall be loaded in accordance with DOT regulations in a tank car conforming to DOT regulations. At the time of filling, the interior of the tank car shall be clean and dry. The tank car shall be closed in a manner to assure purity and integrity of contents from supply source to destination.

5.2 Marking. Each container shall be marked in accordance with DOT regulations and shall as a minimum be marked to show the net weight of contents, lot number, name of manufacturer, date of manufacture, and percent purity. In addition, each container shall be marked: "KEEP DRY - Methylphosphonic Dichloride, Technical."

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Unit quantity required (see 5.1).
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

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(3) Time required for the Government to notify the supplier whether or not to proceed with production.
(4) Size of preproduction lot (see 4.3.1).
(5) Sampling of the preproduction lot (see 4.3.2).

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.3 Sampling and testing precautions. Methylphosphonic dichloride is extremely corrosive and toxic, and reacts violently with water. Extreme care should be exercised at all times to avoid any contact of the material with water. Also, avoid inhalation and contact with skin, eyes, and clothing.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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APPENDIX

SAMPLE PURITY BY NMR CALCULATIONS

10. SCOPE

10.1 This appendix contains sample mole percent purity by NMR calculations and is not a mandatory part of the specification. The information contained herein is intended for guidance only. The numbers used in the calculations do not represent expected test results.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. CALCULATIONS

Step 1. Assign resonances in the ^{31}P NMR spectrum (see figure 4).

	$CH_3P(O)(Cl)_2$	delta 46.6
(Trimer)	$CH_3P(O)OP(O)OP(O)CH_3$ with Cl, CH_3, Cl	delta 33 and delta 14.6
(Dimer)	$CH_3P(O)OP(O)CH_3$ with Cl, Cl	delta 32.6
	Unknown CH_3-P(O)	delta 60.6
	$CH_3P(O)(OH)(Cl)$	delta 41.0

Step 2. Add areas of impurities containing CH_3-P moiety (see figure 5).

	1.8	(41.0)
	3.3	(14.6)
	6.5	(33)
	11.1	(32.6)
+	0.5	(60.6)
	23.2	

(Values are all on the same scale.)

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Step 3. Calculate area percent of methylphosphonic dichloride (see figure 3).

$$\text{Area percent} = \frac{350.2}{398.5} \times 100 \quad \begin{matrix}\text{(integral of methylphosphonic dichloride)} \\ \text{(total integral minus sidebands)}\end{matrix}$$

$$= 87.9 \text{ percent}$$

Step 4. Calculate the integral value/proton for methylphosphonic dichloride (see figure 2).

$$\frac{218.0 \times 0.879}{3} = 63.87$$

Step 5. Calculate the area percent of CH_3-P impurities (see figure 2).

63.87 x 3 = 191.6 (area of methylphosphonic dichloride)

218.0 (total integral)
- 191.6 (integral of methylphosphonic dichloride)

26.4 (area of CH_3-P impurities)

Step 6. Calculate the integral value/proton for each CH_3-P impurity.

$CH_3P(O)(OH)(Cl)$ $\frac{1.8}{23.2} \times \frac{26.4}{3} = 0.68$

Trimer $\frac{9.8}{23.2} \times \frac{26.4}{9} = 1.24$

Dimer $\frac{11.1}{23.2} \times \frac{26.4}{6} = 2.11$

Unknown CH_3P $\frac{0.5}{23.2} \times \frac{26.4}{3} = 0.19$

Step 7. No other phosphorus-containing impurities observed (see figure 4).

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Step 8. Calculate integral values for proton-containing impurities (see figure 2).

Exchangeable proton peak (mostly HCl) delta 5.8

$$\frac{20.00}{1.6(32)} = \frac{X}{1.6(16)}$$

X = 10.00 (total exchangeable proton integral, scaled)

10.00	(total exchangeable proton integral)
- 0.68	(integral for -OH of $CH_3P(O)(OH)(Cl)$)
9.32	(integral value/proton for HCl)

Step 9. Sum all integral values.

Methylphosphonic Dichloride	63.87
HCl	9.32
Dimer	2.11
Trimer	1.24
$CH_3P(O)(OH)(Cl)$	0.68
Other CH_3	+ 0.19
	77.41

Step 10. Calculate mole percent purity of methylphosphonic dichloride.

$$\frac{63.87}{77.41} \times 100 = 82.5 \text{ mole percent}$$

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Step 11. OPTIONAL. Calculate mole percent of each impurity present.

HCl	$\frac{9.32}{77.41} \times 100 = 12.0$ mole percent
Dimer	$\frac{2.11}{77.41} \times 100 = 2.7$ mole percent
Trimer	$\frac{1.24}{77.41} \times 100 = 1.6$ mole percent
$CH_3P(O)(OH)(Cl)$	$\frac{0.68}{77.41} \times 100 = 0.9$ mole percent
Other CH_3	$\frac{0.19}{77.41} \times 100 = 0.2$ mole percent

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